

158.888 Information Technology Professional Project

**Reflection and Peer Review Report on
Real-Time Parking Occupancy Monitoring and Area-
Level Prediction Using Spatio-Temporal Analytics**

Presented by
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1. Reflection on the Project

This project improved my understanding of collaborative system development, transportation analytics, and dashboard-based decision support systems. I gained practical experience in integrating multiple technologies, analysing transportation datasets, and contributing to the development of predictive analytics solutions for smart parking management. The project also enhanced my knowledge of intelligent transportation systems and data-driven urban mobility analysis.

One of the strongest aspects of the project was the development of operational Power BI dashboards capable of visualizing occupancy forecasting, parking utilization trends, hotspot analysis, and smart parking KPIs. Through dashboard development activities, I gained valuable experience in designing analytical interfaces that support real-time operational decision-making. The dashboard implementation helped me understand how data visualization techniques can transform analytical outputs into meaningful operational insights for transportation management systems.

The project involved several practical challenges related to visualization consistency, forecasting integration, and dashboard usability. Integrating machine learning outputs, occupancy categorization, GIS-based hotspot maps, and forecasting visuals required multiple rounds of dashboard refinement and testing. Maintaining consistency across multiple dashboard pages, KPI cards, and forecasting visuals also required careful coordination and iterative improvements.

Another important learning experience involved understanding the relationship between machine learning forecasting outputs and operational dashboard presentation. The project demonstrated that accurate forecasting alone is not sufficient unless the outputs can be effectively visualized and interpreted through user-friendly reporting systems. This significantly improved my understanding of dashboard storytelling and operational analytics presentation.

The project also provided valuable experience in integrating multiple analytical components within a single operational framework. Combining machine learning forecasting, GIS-based spatial analytics, live API integration, and Power BI dashboards required collaborative coordination and continuous testing. Participating in these integration activities improved my understanding of interdisciplinary smart city analytics and intelligent transportation systems.

One of the most challenging parts of the project involved dashboard customization and occupancy categorization. Different visualization styles, forecasting outputs, and occupancy distributions needed to be synchronized carefully to ensure consistency across operational dashboards. Several rounds of debugging and refinement were required to align occupancy categories, hotspot indicators, KPI cards, and forecasting visuals correctly.

The project also strengthened my collaborative communication and project coordination skills. Frequent communication between forecasting development, GIS analysis, and visualization workflows helped ensure successful integration of all system components into the final smart parking intelligence platform. Working in a collaborative technical environment improved my ability to coordinate implementation tasks, testing activities, and reporting workflows effectively.

If I were to repeat the project again, I would allocate more time to early dashboard architecture planning and visualization standardization. Defining dashboard templates, KPI structures, and occupancy categorization frameworks earlier in the project lifecycle could help reduce later modifications and improve implementation efficiency.

Overall, the project provided highly valuable practical experience in dashboard development, transportation analytics, smart city reporting systems, and operational data visualization. It strengthened both my technical and collaborative skills while improving my understanding of how predictive analytics and dashboard intelligence can support smart transportation management and urban mobility decision-making.

2. Peer Review

My primary role in the project involved Power BI dashboard development, visualization integration, KPI card implementation, occupancy categorization, and dashboard styling. I mainly worked on interactive reporting, operational dashboard design, and forecasting visualization components. I was also involved in occupancy distribution visualization, dashboard synchronization, forecasting chart refinement, and operational KPI implementation.

Throughout the project, I contributed significantly to improving the visual presentation and operational usability of the smart parking intelligence platform. This included designing dashboard layouts, implementing forecasting visuals, refining hotspot visualizations, customizing operational KPI cards, and ensuring that occupancy forecasting outputs were effectively presented through interactive dashboards.

S M Amit Hasan mainly handled machine learning forecasting implementation, feature engineering, prediction validation, and API-based forecasting integration. Their contribution formed the analytical and predictive backbone of the project. They were heavily involved in machine learning experimentation, occupancy forecasting optimization, feature engineering workflows, and real-time prediction integration activities.

Amit Chakrabarty mainly contributed to GIS-based spatial analysis, hotspot visualization, documentation support, and dashboard testing activities. Their work helped improve the spatial intelligence and geographic visualization quality of the project. They also supported occupancy categorization review, dashboard testing, and documentation refinement throughout the project lifecycle.

The project tasks were distributed according to individual technical expertise and collaborative coordination throughout the implementation lifecycle. Frequent communication and collaborative testing sessions helped ensure successful integration between machine learning outputs, dashboard visualizations, GIS mapping, and operational reporting systems.

One of the major strengths of the team collaboration was the ability to combine multiple technical domains into a unified operational framework. Since the project integrated forecasting analytics, GIS visualization, dashboard development, and transportation analytics, effective coordination between team members became essential. Regular discussions and collaborative debugging sessions helped resolve implementation challenges and improve the consistency of the final dashboards and forecasting outputs.

Overall, the teamwork structure was highly effective and contributed significantly to the successful completion of the project. Each member contributed according to their specialization while also supporting testing, refinement, and integration activities across different implementation stages.